

On Chian Yee

Data Science Master's Student | Machine Learning & Analytics | [Portfolio](#)
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Education

Monash University Malaysia

Master of Data Science | CGPA: 3.63

Bandar Sunway, Selangor

Mar 2026 – June 2027 (Expected)

- Relevant Coursework: Machine Learning, Project Management, Statistical Modelling
Deep Learning, Artificial Intelligence, Data Wrangling

Bachelor of Computer Science in Data Science | CGPA: 3.09

Oct 2022 – Nov 2025

- Relevant Coursework: Data Modelling, Data Analytics, Data Visualisation, Algorithms & Data Structures
Big Data Management and Processing, Object-Oriented Programming

Study Abroad

Study abroad coursework at Monash University Clayton campus.

Melbourne, Victoria

Feb 2024 – June 2024

Skills & Interests

Programming: Python (pandas, NumPy, scikit-learn, TensorFlow/Keras), R, SQL, Java, JavaScript

Data & Analytics: Tableau, Excel, Spark, Kafka, MongoDB, SAS Viya, SPSS (basic)

Deployment & Tools: Flask, Docker, Heroku, Git/GitHub, AWS (fundamentals), HTML/CSS

Language: Mandarin, English, Bahasa Malaysia, Cantonese, Hakka

Certificates: [TeaMWork](#), [Understanding Digital Transformation](#)

Project Experience – Explore in [My Portfolio](#)

Multi-Agent Train Scheduling under Malfunctions | [2026 Portfolio](#)

Sept 2026

- Built conflict-free route planners in Python for the Flatland railway simulator, progressing from single-train A* search to multi-agent planning with space-time reservation tables.
- Implemented Safe Interval Path Planning, prioritised planning and Large Neighbourhood Search with malfunction-aware replanning, plus cached reverse-Dijkstra heuristics to cut search time.

Machine Learning Algorithms from Scratch | [2026 Portfolio](#)

May 2026

- Derived and implemented soft- and hard-EM for multinomial mixture models in NumPy (log-space) to cluster news articles over a 30K-word vocabulary, comparing convergence speed and solution quality.
- Compared L2-regularised Perceptrons with 3-layer neural networks on non-linear data, and built a PyTorch autoencoder for self-taught digit classification with only 40 labelled samples.

Multi-omics machine learning model for predicting chemoresistance | [2025 Portfolio](#)

August 2024 – June 2025

- Built a deep neural network (TensorFlow/Keras) integrating transcriptomic (19K+ features), proteomic and drug-structure (RDKit/isoSMILES) data across 224 cancer cell lines.
- Ran systematic experiments across 5+ combinations of data types with grid-searched architectures; showed that adding genomics reduced accuracy and removed it, which also simplified preprocessing.
- Led architecture design and deployment of a Flask web app (Docker, Heroku) giving real-time drug-response predictions from uploaded CSVs, with top-10 drug visualisations.

Work Experience & Activities

Persatuan Komputer Universiti Malaya (PEKOM)

UMHackathon 2025

Online

April 2025 – April 2025

- Designed a chat-based AI assistant prototype for Grab merchants that delivers real-time business insights, personalised recommendations, and multilingual communication support to enhance operational decision-making.

IPSOS Sdn Bhd

Internship

Mid Valley City, Kuala Lumpur

November 2024 – February 2025

- Collected, cleaned and validated survey data for customer experience research projects, keeping datasets accurate and client-ready.
- Analysed results with Excel formulas and built charts and PowerPoint analysis reports for project teams.